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Journal of Economic Psychology 22 (2001) 377–395

JOURNAL OF
**Economic
Psychology**

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Brand personality: How to make the metaphor fit? ☆

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Received 2 July 1997; received in revised form 6 May 1998; accepted 1 January 2001

Abstract

The Big Five Model of human personality [Goldberg, 1990, *J. Pers. Soc. Psychol.* 59, 1216] reduces the large number of adjectives describing human personalities to only five latent dimensions (the so-called *Big Five* Factors of Extroversion, Agreeableness, Conscientiousness, Emotional Stability, and Openness), which provide a consensual framework for classifying and organizing descriptors of human personality. This paper examines 12 mass-market brands to determine to what extent, in a consumer setting, the Big Five can serve as a metaphor to describe enduring characteristics of brands. More than 1500 subjects evaluated their own personalities and those of three brands by using 40 adjectives (8 for each trait) typical of the dimensions of human personality according to the Big Five Factor Model. Results from exploratory factor analyses showed that the five-factor structure is not replicated when describing brands. Rather, at a higher level of abstraction in the hierarchical organization of personality characteristics, results supported a two-trait solution. It was also found that descriptors of human personality convey different meanings when attributed to different brands.

☆ This paper was produced while Prof. G.V. Caprara was a Golestan Fellow-in-Residence at the Netherlands Institute for Advanced Study in the Humanities and Social Sciences, The Netherlands, and Prof. G. Guido was visiting researcher at the Department of Marketing of the University of Florida at Gainesville.

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While the psycholexical approach remains a suitable procedure to identify brand descriptors, the factors used to describe human personalities appear to be inappropriate for describing the brands studied here. © 2001 Elsevier Science B.V. All rights reserved.

PsycINFO classification: 3900

JEL classification: D12

Keywords: Brand personality; Psycholexical approach; Five factor model

1. Introduction

The applicability of a number of cognitive mechanisms associated with the perception of social stimuli (mainly person-related) to the perception of marketing stimuli (mainly object-related) has been documented in past studies (Aaker, 1997; Fournier, 1995; Heckler & Childers, 1992; Sujan & Bettman, 1989). This makes plausible the hypothesis that through their interactions with products and advertising, consumers come to personify brands as having personalities.

The psycholexical tradition in personality psychology maintains that over time, languages develop a list of adjectives that describe the most important personality differences between people (cf. Goldberg, 1982). Factor analyses among thousands of personality attributes have revealed that, in many different languages, most adjectives describing individuals can be grouped into one of the so-called *Big Five* personality traits of Extroversion, Agreeableness, Conscientiousness, Emotional Stability, and Openness (see Appendix A). It is still questionable, however, whether these Big Five personality traits can be applied to brands (Caprara, Barbaranelli, & Guido, 1998). This paper presents data suggesting that while the notion of brand personality may be useful, the Big Five personality factors are not a good way of thinking about brands.

2. The use of human personality models in the study of brand images

A number of studies have provided empirical support for the hypothesis that personality may be a viable metaphor for understanding consumers' perception of brand images. In an attempt to provide an unequivocal interpretation of the very unstable concept of brand image, a consolidated

research tradition has put the principal emphasis on personification of brands (cf. Dobni & Zinkhan, 1990). However, this tradition has suffered due to the lack of both a common theory and a consensual taxonomy of personality traits. This has led to a proliferation of constructs, severely undermining the comparability of different approaches and even the credibility of the construct itself. Biehl (1993), for example, suggested that brand image is composed of the image of the provider (e.g., the manufacturer), the user, and the product itself. Farquhar and Herr (1993) argued that the associations evoked by brands include specific product attributes, customer benefits, usage situations and other summary evaluations. Coulter and Zaltman (1994) and Fournier (1995, 1998), using in-depth interviews, maintained that brands have meaning in themselves, and add meaning to a person's life, through their status as partners in a relationship with the consumer. Recently, a consensus has emerged among personality psychologists around the Big Five Model as a reference structure for the assessment and description of human personality, but applications of this model to marketing settings have appeared only recently (Aaker, 1995, 1997; Aaker & Fournier, 1995; Caprara & Barbaranelli, 1996; Caprara et al., 1998).

2.1. The lexical approach to the study of personality and the Big Five Model

The Big Five Model emerged in studies that examined the “language” of personality within the framework of the *psycholexical* approach. This approach originated from a hypothesis, formulated by Gordon Allport at the end of the 1930s and formalized by Raymond Cattell in the mid-1940s, as “linguistic sedimentation”, or the “lexical hypothesis”. According to this approach, nouns and adjectives that describe human personality are integral to the development and maintenance of social relations. As such, they become part of the vocabulary used by people every day, and are transmitted from one generation to another through processes of socialization. The practical consequence is that the vocabulary of natural languages represents the main source of descriptors of personality characteristics (cf. John, Angleitner, & Ostendorf, 1988). Several studies, scanning thousands of adjectives and nouns in unabridged dictionaries of different languages (cf. Saucier, Hampson, & Goldberg, 1999), selected terms denoting stable characteristics of human personalities, which have been mostly referred as the least ambiguous, the most frequently used, and the most useful for human personality description.

Surprisingly, factor analysis conducted on these data sets revealed a structure generally composed of only five broad personality dimensions/traits (the so-called “Big Five Factors”). They are: (1) *Extroversion*, the preference for social interaction and for activity; (2) *Agreeableness*, the orientation toward compassion and caring about others, and away from antagonism; (3) *Conscientiousness*, the preference for goal-oriented activity (i.e., the degree of organization); (4) *Emotional Stability*, the ability to cope effectively with negative emotions; and (5) *Openness to Experience*, the tolerance for new ideas and new ways of doing things, experientially oriented (see, for complete reviews of these studies, Digman, 1990; Goldberg, 1993; McCrae & John, 1992; Wiggins & Pincus, 1992). These traits were found to be easily evoked by a limited number of adjectives (so-called *markers*) that showed a high loading in the desired factor and low loadings in the remaining factors (Goldberg, 1992).

Further confirmations of this five-factor structure came from studies of personality traits using self-report questionnaires and others’ ratings (see, for a review, Ostendorf & Angleitner, 1992). Most recently, by re-analyzing factor solutions derived from human personality descriptions, Digman (1997) found second-order dimensions that account for correlations among the Big Five, resulting in two higher-level factors. According to Digman, these two factors represent the highest level of a hierarchical model of personality characteristics in which personality terms are arranged in order of increasing complexity, from the most specific behavioral response to the broader personality dimensions. The first dimension (named *Factor α*) is defined by the common aspects of Agreeableness, Conscientiousness and Emotional Stability and represents such personality aspects as socialization, social interest, union, communion, and intimacy. The second dimension (named *Factor β*) is defined by the common aspects of Extroversion and Openness to Experience and represents such personality aspects as striving for superiority, individuation, personal growth, achievement of status, and desire for power. These two factors can, thus, be considered as meta-traits at a higher level of abstraction, which do not invalidate the soundness of the five-factor structure.

2.2. *Capitalizing on the psycholexical approach and on the Big Five in construing brand personality*

Since brands, like persons, are usually described with adjectives (see Berry, 1988; Durgee, 1988; Plummer, 1984/85; Poiesz, 1989), the psycholexical ap-

proach seems to be a good method for identifying the main characteristics of brands' personalities in the perception of consumers, and to select the best adjectives for conveying certain characteristics. In reality, it remains questionable whether the principal features of brands (even the well-established ones) can be encoded as stable traits and expressed by single words, as seems to be the case with human traits. A recent study by Aaker (1997), focused on a general cross-category framework, investigated brand personality dimensions underlying 114 traits used to describe 37 different brands. After sorting a number of descriptors of brand personality, a five-factor solution was found. This was corroborated by both exploratory and confirmatory factor analyses, as well as by replicability analyses. Yet only three of these factors were those appearing in the Big Five Model (see, in particular, Aaker, 1997, Appendix A).

Even when the personality metaphor seems suitable for brands, marketers interested in shaping and reinforcing brands' desirable features need to know whether the same adjectives correspond to the same factors when used to describe personalities of different brands. It is important not only to ascertain the applicability to brands of those traits and markers that proved valid to describe humans, but also to select those traits and markers that fit best with the brand personality that the marketer intends to establish or reinforce. Therefore, the focus of this paper was the investigation of the different meanings conveyed by trait names when associated to specific brands. By using the Big Five Model as a framework, this study examined the personality of 12 brands from different market segments using a list of adjectives commonly used to describe human personality. Our aim was threefold. We sought to determine: (a) whether the Big Five Model of human personality is useful for the description of brand personality; (b) whether markers of human personality applied to brand personality are traceable to the same factor solution found in humans; and (c) whether personality descriptors load under the same factor when used to describe human personality and brand personalities, and when used to describe the personalities of different brands.

3. Method

3.1. Subjects

The participants were 1586 adult Italian subjects (791 males, 783 females, 12 unreported gender), aged from 20 to 65 years (mean = 41; S.D. = 14.11).

They were recruited by undergraduate students, as a requirement for a personality psychology course. Each student was asked to collect six subjects according to the research design (i.e., 1 male and 1 female in the 20–35 year-old range; 1 male and 1 female in the 36–50 year-old range; and 1 male and 1 female in the 51–65 year-old range).

3.2. Target stimuli

Twelve different brands in the Italian market were chosen from a pool of brands well known to all subjects as belonging to leading companies in different business fields. They included: AGIP (a major Italian petrol producer), Alitalia, Barilla (a prominent Italian pasta and food manufacturer), Benetton, BNL (a notable Italian bank), Coca-Cola, Ferrero (a renowned Italian confectioner), Fiat, IBM, Mondadori (the largest Italian publishing company), Olivetti, and Sony. Participants' knowledge of the brands was ascertained by a short questionnaire in which they were asked whether they knew the brands, and if they had ever used them. Only those participants who reported knowing the brand were considered in the study.

3.3. Research design

Subjects were asked to describe first the perceived personality of three brands (randomly selected out of the above-mentioned 12 brands), then their own personality. This was done using a list of 40 adjectives comprising eight markers of each of the Big Five (see Appendix A). These markers were taken from a wider list of about 500 trait terms identified as the most useful for describing personality in the Italian language (see Caprara & Perugini, 1994). Each adjective was rated on a 5-point Likert scale.

3.4. Statistical analysis

Principal component analyses were performed to analyze the evaluation of the subjects' own personalities and those of the 12 brands. In each analysis, the Kaiser–Meyer–Olkin (KMO) test and Bartlett's test of sphericity were used to assess the appropriateness of the correlation matrices to factor analysis. After having ascertained the suitability of the correlation matrices to factor analysis, a scree test of eigenvalues was used to select the number of components to be included in the rotation process. Once the number of

factors to retain was decided, Oblimin oblique rotations were applied to unrotated matrices.

Replicability analysis was used to compare the expected five factors with the empirical results of the exploratory factor analysis. The core issue of this analysis is that only those factors that are replicable can be considered meaningful. Replicability analysis was performed using Everett's method (Everett, 1983), which has been the focus of several recent contributions aimed at examining the structure of personality within the domain of the Big Five Model (e.g., Borkenau & Ostendorf, 1990; McCrae & Costa, 1987). For each analysis a given number of factors was extracted separately for any two brands, and, after oblique rotation, factor scoring coefficients were computed. These scoring coefficients were applied to standardized scores of personality descriptions of the two brands to generate two sets of factor scores, which were then correlated. For the sake of simplicity, replicability coefficients were averaged across all 12 brands, to obtain only one coefficient for each factor in each solution.

4. Results

4.1. *Exploratory factor analysis*

Results of both the KMO and Bartlett's tests applied to the 13 correlation matrices clearly demonstrated the factorability of the matrices considered. Table 1 shows the eigenvalues relative to the first 10 extracted factors in the 13 different analyses, and the percentage of variance explained by the unrotated factors. Inspection of these values immediately shows clear differences between the factor solutions emerging from brands' personalities and from individuals' personality descriptions. In the case of the analysis performed on the descriptions of brands' personalities, there is a consistent drop in magnitude after the second eigenvalue and a consistent decline in explained variance after the second factor. Conversely, in the case of the analysis performed on the descriptions of subjects' own personalities, the curve described by the eigenvalues is less steep and the variance explained by the factors is more spread. While the first *two* extracted factors of brand personalities explain on the average 45% of the variance (ranging from 41.12% of Benetton, to 52.30% of Olivetti), about the same amount of variance is explained by the first *five* extracted factors of human personality. This attests to a restriction of

Table 1

Eigenvalues (first row) and percentage of Variance Explained (second row) relative to the factor analyses on brand and human personality

	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	
BNL	13.60	3.60	2.13	1.65	1.42	1.22	1.08	1.02	0.91	0.85	
	33.99	9.01	5.32	4.13	3.55	3.05	2.70	2.55	2.27	2.11	42.99 ^a
BENETTON	12.59	3.85	2.13	1.59	1.49	1.13	1.08	0.98	0.94	0.90	
	31.48	9.64	5.33	3.97	3.73	2.82	2.70	2.45	2.35	2.25	41.12 ^a
AGIP	14.91	4.07	1.58	1.48	1.19	1.15	1.06	0.98	0.89	0.81	
	37.26	10.18	3.95	3.69	2.99	2.86	2.65	2.44	2.22	2.03	47.44 ^a
OLIVETTI	15.59	5.33	1.44	1.34	1.06	1.00	0.88	0.81	0.76	0.71	
	38.96	13.33	3.59	3.34	2.66	2.51	2.19	2.02	1.91	1.77	52.30 ^a
SONY	12.96	4.62	1.80	1.47	1.18	1.15	0.94	0.90	0.86	0.84	
	32.39	11.56	4.50	3.67	2.95	2.87	2.35	2.25	2.16	2.11	43.95 ^a
FERRERO	13.46	3.09	1.87	1.72	1.24	1.09	1.07	0.99	0.96	0.88	
	33.66	7.73	4.68	4.30	3.09	2.71	2.68	2.48	2.41	2.20	41.39 ^a
FIAT	14.87	4.54	1.80	1.45	1.14	1.05	0.99	0.91	0.87	0.79	
	37.19	11.34	4.50	3.63	2.86	2.62	2.47	2.28	2.17	1.96	48.53 ^a
MONDADORI	12.16	4.63	2.14	1.69	1.47	1.19	1.05	1.03	0.94	0.85	
	30.41	11.58	5.35	4.23	3.67	2.99	2.62	2.58	2.35	2.13	41.99 ^a
BARILLA	14.44	3.69	1.64	1.46	1.24	1.13	1.02	1.00	0.91	0.89	
	36.09	9.23	4.09	3.65	3.09	2.84	2.55	2.51	2.27	2.23	45.31 ^a
COCA-COLA	12.63	5.69	1.52	1.33	1.18	1.15	1.09	0.95	0.93	0.84	
	31.58	14.22	3.81	3.31	2.94	2.88	2.72	2.38	2.31	2.10	45.79 ^a
ALITALIA	14.62	3.76	1.73	1.57	1.22	1.04	0.98	0.95	0.90	0.84	
	36.54	9.41	4.33	3.93	3.05	2.61	2.44	2.37	2.25	2.11	45.94 ^a
IBM	12.52	4.77	1.97	1.49	1.38	1.15	1.01	0.96	0.93	0.87	
	31.31	11.91	4.92	3.73	3.45	2.86	2.52	2.39	2.33	2.17	43.22 ^a
SELF-REPORT	7.87	3.66	3.03	2.38	1.63	1.33	1.16	1.07	0.98	0.97	
	19.90	9.20	7.60	6.00	4.10	3.30	2.90	2.70	2.40	2.40	46.40 ^b

^a Cumulated percentage of variance explained by the first two factors.

^b Cumulated percentage of variance explained by the first five factors.

Table 2

Summary of results of factor analyses on human and brand personality^a

	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5
BNL	ES/A	O/CE	–	–	–
BENETTON	ES/A/C	O/E	–	–	–
AGIP	ES/A	O/E	–	–	–
OLIVETTI	ES/A/(E)	O/(E)/C	–	–	–
SONY	ES/A/(E)	O/(E)/C	–	–	–
FERRERO	ES/A/(C)	O/E/(C)	–	–	–
FIAT	O/E/C	ES/A	–	–	–
MONDADORI	ES/A	O/E/C	–	–	–
BARILLA	ES/A	O/E/C	–	–	–
COCA-COLA	ES/A/C	O/E	–	–	–
ALITALIA	O/E/C	ES/A	–	–	–
IBM	ES/A	O/(E)/C	–	–	–
Self-report	ES	O	E	A	C

^a ES = Emotional Susceptibility; O = Openness; E = Energy; A = Agreeableness; C = Conscientiousness.

the number of factors in the case of brands or, in other words, a higher discrimination of subjects when evaluating their own personalities than when evaluating brand personalities.

Based on these results, two factors and five factors, respectively, were rotated for brands and for human personality. Table 2 summarizes the principal findings of these 13 analyses, and assigns a label to each factor on the basis of the rotated pattern matrices.

4.2. Brand personality

Table 3 presents the rotated factor pattern of the 12 principal component analyses performed on brand personalities. One of the two factors that emerged was identified by the markers of Emotional Stability and Agreeableness and was found in all 12 analyses. The other factor was identified, respectively, by markers of Openness, Conscientiousness, and Extroversion in eight cases; Openness and Extroversion in three cases; and Openness and Conscientiousness in one case. Solutions with higher number of factors were attempted, but none resulted in a factor pattern clearer than the one resulting from the 2-factor solution. The two factors showed significant correlations, ranging from 0.29 (for Coca-Cola) to 0.49 (for Alitalia), with an average of 0.40 (S.D. = 0.07).

To examine the similarities among the two factors identified in the analysis of brand personality, Tucker's phi congruence coefficients were

Table 3

Results of exploratory factor analyses of brand personality

	BNL		Benetton		Agip		Olivetti		Sony		Ferrero	
	F1	F2	F1	F2	F1	F2	F1	F2	F1	F2	F1	F2
Active	-0.03	0.68	0.15	-0.49	0.05	0.66	0.17	0.64	0.30	0.45	0.19	0.47
Affectionate	0.71	0.04	0.69	0.00	0.68	-0.07	0.80	-0.12	0.76	-0.22	0.75	-0.08
Altruist	0.53	0.09	0.52	0.01	0.50	0.17	0.59	0.03	0.60	-0.03	0.59	0.04
Authentic	0.44	0.32	0.31	-0.42	0.53	0.19	0.26	0.53	0.10	0.51	0.15	0.51
Calm	0.56	-0.02	0.75	0.30	0.79	-0.24	0.82	-0.14	0.77	-0.21	0.73	-0.16
Competitive	-0.07	0.74	-0.04	-0.57	-0.24	0.76	-0.10	0.79	-0.18	0.62	0.02	0.47
Conscientious	0.26	0.34	0.61	-0.02	0.25	0.43	0.50	0.29	0.45	0.27	0.31	0.40
Constant	0.27	0.42	0.47	-0.27	0.47	0.33	0.39	0.45	0.43	0.37	0.28	0.38
Cordial	0.43	0.35	0.65	-0.08	0.62	0.13	0.78	-0.04	0.74	0.02	0.65	0.14
Creative	0.55	0.26	-0.10	-0.77	0.30	0.39	0.23	0.58	0.14	0.58	0.15	0.64
Dominant	-0.06	0.35	-0.11	-0.51	0.00	0.42	0.16	0.47	0.06	0.48	-0.10	0.52
Efficient	0.09	0.64	0.24	-0.41	-0.03	0.76	-0.01	0.72	-0.02	0.61	0.21	0.51
Energetic	0.31	0.34	0.15	-0.46	0.36	0.51	0.39	0.42	0.56	0.29	0.35	0.41
Faithful	0.47	0.32	0.56	-0.32	0.64	0.19	0.50	0.35	0.47	0.36	0.50	0.16
Fanciful	0.65	0.04	-0.13	-0.76	0.52	0.26	0.42	0.31	0.42	0.42	0.16	0.65
Generous	0.62	0.02	0.54	-0.04	0.66	0.09	0.80	-0.02	0.72	-0.04	0.60	0.12
Genuine	0.74	-0.06	0.60	-0.20	0.63	0.10	0.65	0.18	0.49	0.27	0.22	0.29
Happy	0.73	-0.01	0.21	-0.40	0.64	0.04	0.74	-0.08	0.63	-0.01	0.41	0.32
Informed	-0.15	0.68	0.12	-0.59	0.12	0.54	-0.10	0.77	0.27	0.51	0.13	0.56
Innovating	0.31	0.43	-0.01	-0.78	0.06	0.62	-0.15	0.79	-0.07	0.71	-0.21	0.84
Level-headed	0.46	0.28	0.67	-0.02	0.54	0.26	0.49	0.36	0.57	0.21	0.35	0.44
Light-hearted	0.75	-0.17	0.42	-0.20	0.88	-0.18	0.84	-0.14	0.68	-0.06	0.58	0.01
Lively	0.65	0.03	0.09	-0.64	0.65	0.22	0.76	0.05	0.61	0.16	0.49	0.22
Loyal	0.28	0.47	0.68	-0.10	0.60	0.28	0.64	0.19	0.63	0.21	0.67	0.13
Modern	0.04	0.63	-0.02	-0.69	-0.04	0.68	-0.21	0.79	-0.22	0.73	-0.21	0.76
Original	0.65	0.09	0.04	-0.62	0.60	0.26	0.29	0.39	0.11	0.53	0.08	0.53
Patient	0.75	0.04	0.76	0.06	0.80	-0.09	0.84	-0.03	0.81	-0.12	0.77	-0.03
Precise	0.00	0.69	0.49	-0.28	0.40	0.45	0.09	0.65	0.12	0.57	0.52	0.26
Productive	-0.13	0.80	0.07	-0.58	-0.13	0.72	-0.11	0.72	0.17	0.56	0.01	0.56
Recent	0.01	0.70	-0.08	-0.69	-0.04	0.70	-0.14	0.78	-0.19	0.65	-0.16	0.76
Regular	0.17	0.54	0.55	-0.01	0.45	0.33	0.55	0.34	0.51	0.34	0.58	0.07
Relaxed	0.73	-0.21	0.57	-0.02	0.87	-0.23	0.88	-0.21	0.80	-0.17	0.77	-0.21
Reliable	0.07	0.62	0.40	-0.38	-0.01	0.58	0.08	0.65	-0.07	0.56	0.25	0.37
Resolute	0.20	0.52	0.10	-0.60	0.33	0.48	0.44	0.44	0.39	0.45	0.33	0.41
Scrupulous	0.08	0.63	0.54	-0.17	0.43	0.37	0.43	0.38	0.40	0.40	0.40	0.33
Serene	0.74	0.04	0.67	-0.08	0.87	-0.10	0.84	0.00	0.77	0.00	0.83	-0.06
Stable	0.05	0.58	0.44	-0.20	0.26	0.45	0.35	0.44	0.38	0.40	0.48	0.12
Strong	-0.03	0.51	0.15	-0.45	0.22	0.53	0.28	0.49	0.33	0.50	0.35	0.34
Tranquil	0.70	-0.02	0.78	0.31	0.83	-0.19	0.87	-0.11	0.79	-0.15	0.73	-0.09
Up-to-date	-0.06	0.69	-0.03	-0.75	-0.03	0.75	-0.11	0.83	-0.18	0.70	-0.01	0.65
% of explained variance	22	21	21	20	28	20	29	24	25	19	22	19

Table 3 (Continued)

	Fiat		Mondadori		Barilla		Coca-Cola		Alitalia		IBM	
	F1	F2	F1	F2	F1	F2	F1	F2	F1	F2	F1	F2
Active	0.71	-0.10	0.06	0.58	-0.23	0.54	0.34	0.51	0.72	0.00	0.05	0.64
Affectionate	0.04	0.70	0.69	-0.12	-0.71	-0.03	0.22	0.46	-0.04	-0.72	0.69	-0.24
Altruist	0.09	0.55	0.56	0.02	-0.53	0.10	0.41	0.25	0.06	-0.52	0.42	0.04
Authentic	0.41	0.25	0.33	0.40	-0.14	0.57	-0.18	0.68	0.41	-0.31	0.23	0.33
Calm	-0.17	0.77	0.69	-0.09	-0.80	-0.13	0.79	-0.19	-0.01	-0.68	0.63	0.00
Competitive	0.84	-0.19	-0.31	0.67	0.17	0.52	-0.22	0.72	0.74	0.07	-0.11	0.64
Conscientious	0.45	0.32	0.27	0.47	-0.28	0.41	0.70	0.01	0.50	-0.17	0.37	0.25
Constant	0.41	0.22	0.41	0.31	-0.26	0.43	0.21	0.57	0.49	-0.32	0.37	0.44
Cordial	0.14	0.72	0.65	0.12	-0.73	0.06	0.19	0.62	0.25	-0.46	0.65	-0.06
Creative	0.68	0.10	0.13	0.60	-0.11	0.60	-0.03	0.72	0.19	-0.52	0.21	0.42
Dominant	0.70	-0.15	-0.11	0.38	0.05	0.61	0.37	0.33	0.51	0.02	0.00	0.51
Efficient	0.72	0.05	0.04	0.73	-0.22	0.53	0.57	0.22	0.85	0.14	0.01	0.76
Energetic	0.67	0.16	0.29	0.45	-0.34	0.42	0.20	0.53	0.43	-0.31	0.51	0.26
Faithful	0.25	0.48	0.58	0.21	-0.61	0.18	0.08	0.66	0.29	-0.54	0.56	0.21
Fanciful	0.58	0.17	0.16	0.58	-0.05	0.62	0.05	0.63	0.06	-0.64	0.46	0.17
Generous	0.10	0.72	0.70	0.02	-0.61	0.12	0.74	0.07	0.06	-0.63	0.72	-0.15
Genuine	0.10	0.62	0.55	0.17	0.02	0.52	0.80	-0.26	-0.05	-0.75	0.65	0.05
Happy	0.38	0.39	0.54	0.05	-0.52	0.21	0.23	0.55	-0.04	-0.72	0.66	0.01
Informed	0.58	0.21	-0.08	0.75	-0.26	0.52	0.49	0.28	0.53	-0.15	0.08	0.47
Innovating	0.82	-0.15	0.08	0.63	0.11	0.78	0.44	0.17	0.64	-0.02	-0.07	0.74
Level-headed	0.58	0.16	0.46	0.37	-0.56	0.28	0.77	0.04	0.49	-0.29	0.56	0.22
Light-hearted	0.03	0.70	0.73	-0.22	-0.82	-0.09	0.19	0.59	-0.19	-0.80	0.75	-0.21
Lively	0.35	0.48	0.58	0.16	-0.50	0.34	0.39	0.51	-0.03	-0.76	0.62	0.12
Loyal	0.17	0.67	0.69	0.17	-0.59	0.12	0.89	-0.23	0.24	-0.57	0.69	0.14
Modern	0.77	-0.09	0.06	0.58	0.13	0.70	-0.09	0.47	0.77	0.12	-0.06	0.71
Original	0.57	0.11	0.38	0.41	-0.30	0.43	0.08	0.56	0.18	-0.51	0.41	0.17
Patient	-0.08	0.83	0.80	-0.07	-0.82	-0.12	0.73	0.00	0.03	-0.72	0.72	0.00
Precise	0.61	0.18	0.30	0.54	-0.29	0.46	0.53	0.29	0.64	-0.14	0.12	0.67
Productive	0.59	0.07	-0.07	0.64	0.06	0.65	-0.21	0.77	0.64	-0.04	-0.06	0.73
Recent	0.78	-0.21	-0.21	0.70	0.06	0.73	0.37	0.12	0.78	0.19	-0.16	0.78
Regular	0.32	0.45	0.43	0.25	-0.50	0.28	0.70	0.04	0.59	-0.08	0.36	0.40
Relaxed	-0.10	0.73	0.78	-0.15	-0.87	-0.16	0.77	-0.12	-0.14	-0.83	0.80	-0.25
Reliable	0.61	0.08	-0.01	0.55	0.10	0.56	0.51	0.26	0.60	0.02	0.00	0.72
Resolute	0.70	0.13	0.17	0.55	-0.29	0.56	-0.11	0.78	0.46	-0.30	0.45	0.42
Scrupulous	0.53	0.28	0.24	0.50	-0.30	0.45	0.78	-0.11	0.49	-0.22	0.53	0.29
Serene	-0.11	0.83	0.78	-0.04	-0.81	-0.01	0.68	0.07	0.00	-0.77	0.84	-0.07
Stable	0.57	0.06	0.33	0.34	-0.24	0.43	0.83	-0.19	0.55	-0.25	0.38	0.41
Strong	0.64	0.00	0.11	0.42	-0.21	0.51	0.16	0.58	0.58	-0.11	0.20	0.44
Tranquil	-0.07	0.76	0.75	-0.09	-0.84	-0.09	0.55	0.09	0.08	-0.66	0.73	-0.04
Up-to-date	0.80	-0.18	-0.15	0.72	0.04	0.72	-0.17	0.72	0.71	0.13	-0.11	0.82
% of explained variance	28	21	22	20	23	22	25	21	23	23	24	19

computed for the factor loadings of the 12 analyses. Congruence coefficients ranged from 0.65 to 0.98, $M = 0.90$, $S.D. = 0.03$, in the case of the Extroversion/Openness/(Conscientiousness) factor, and from 0.62 to 0.98, $M = 0.89$, $S.D. = 0.10$, in the case of the Stability/Agreeableness factor. Coefficients of about 0.90 are typically considered to define matching factors (Mulaik, 1972). Although factors' correspondence was very high for all brands (with the sole exception of Coca-Cola), there was a clear tendency for some adjectives to "shift" from one factor (i.e., dimension) to another. Some adjectives appeared under different factors according to the brand they referred to. This phenomenon – probably explained by the influence of brand types on respondents' ratings (see Section 5) – remained despite the imposition of two restrictions: the exclusion of Coca-Cola scores, and the consideration, for each adjective and each brand, only of the differences among two-factor loadings higher than $|0.20|$. Under these conditions for each Big Five dimension, it was possible to identify at least two adjectives that behaved differently according to the brand they were describing.

For the Extroversion dimension, we observed the following "shifting" adjectives: *energetic* (loaded by ES/A factor in the case of Sony and IBM, and by E/A factor in the case of almost all other brands), and *lively* (loaded by E/O/C factor in the case of all brands except for Benetton and Coca-Cola, where it loaded by ES/A factor). For the Agreeableness dimension, we had: *authentic* (loaded by ES/A in the case of BNL and Agip, and by E/O/C in the case of all other brands), and *loyal* (loaded by ES/A factor in the case of all brands except for BNL, where it was loaded by E/O/C factor). For the Conscientiousness dimension, we had: *conscientious* (loaded by ES/A factor in the case of Benetton, Olivetti, Coca-Cola, IBM and Sony, and by E/O factor in the case of the other brands), and *scrupulous* (loaded by ES/A factor in the case of Coca-Cola, IBM, Agip, and Benetton, and by E/O factor in the case of Fiat, Mondadori, Alitalia, Barilla and BNL). For the Emotional Stability dimension, we had: *level-headed* (loaded by ES/A factor in the case of all brands but Alitalia and Fiat, where it loaded by E/O/C factor), and *stable* (loaded by ES/A factor in the case of Ferrero, Coca-Cola and Benetton, and by E/O/C factor in the case of Fiat, Agip, Alitalia, Barilla and BNL). For the Openness dimension, we had: *creative* (loaded by ES/A factor in the case of BNL and Alitalia, and by E/O/C factor in the case of all other brands), and *fanciful* (loaded by ES/A factor in the case of Alitalia, IBM, BNL, Agip and Olivetti, and by O/E/C factor in the case of Benetton, Ferrero, Fiat, Mondadori, Coca-Cola and Barilla).

Table 4
Results of exploratory factor analysis of human personality^a

	ES	O	E	A	C
Tranquil	0.83	−0.02	−0.17	−0.04	0.09
Calm	0.82	−0.03	−0.20	−0.08	0.07
Relaxed	0.79	−0.06	−0.03	−0.13	−0.10
Serene	0.73	0.05	0.18	0.07	−0.11
Patient	0.57	−0.07	−0.15	0.18	0.16
Light-hearted	0.57	−0.10	0.16	0.00	−0.39
Stable	0.42	0.10	0.27	0.10	0.37
Fanciful	−0.01	−0.69	−0.07	0.21	−0.19
Original	0.03	−0.67	0.02	0.13	−0.07
Creative	−0.04	−0.67	0.02	0.14	−0.09
Innovating	0.02	−0.66	0.11	0.05	−0.02
Up-to-date	0.08	−0.63	0.06	−0.17	0.29
Informed	0.09	−0.61	0.05	−0.16	0.28
Recent	0.13	−0.57	0.13	0.02	0.02
Modern	0.04	−0.48	0.20	0.10	−0.02
Energetic	−0.04	−0.05	0.70	0.08	−0.09
Dominant	−0.09	−0.11	0.62	−0.13	−0.06
Active	0.00	−0.05	0.62	0.11	0.05
Strong	0.07	−0.01	0.60	0.05	0.01
Resolute	0.08	−0.16	0.60	−0.02	0.07
Efficient	−0.03	−0.09	0.51	0.09	0.29
Productive	0.01	−0.05	0.50	0.01	0.28
Competitive	0.03	−0.39	0.40	−0.22	−0.03
Generous	0.02	−0.03	0.07	0.71	−0.04
Affectionate	0.11	−0.07	−0.05	0.65	−0.10
Altruist	0.04	−0.01	0.03	0.65	−0.06
Loyal	−0.02	−0.02	0.00	0.54	0.31
Cordial	0.14	−0.11	0.13	0.52	−0.03
Authentic	−0.03	−0.17	0.03	0.48	0.14
Genuine	0.00	−0.15	0.03	0.47	0.13
Faithful	0.01	0.15	−0.02	0.41	0.28
Precise	0.06	−0.13	0.14	−0.03	0.56
Scrupulous	0.00	−0.04	0.10	0.15	0.56
Conscientious	−0.01	−0.03	0.06	0.27	0.53
Lively	0.10	−0.21	0.37	0.35	−0.52
Happy	0.26	−0.13	0.29	0.33	−0.51
Constant	0.28	0.05	0.24	0.07	0.51
Level-headed	0.36	0.03	0.13	0.12	0.43
Regular	0.28	0.11	0.25	0.11	0.43
Reliable	0.08	−0.02	0.07	0.35	0.39
% of explained variance	9.88	9.79	9.48	8.91	8.38

^a ES = Emotional Stability; O = Openness; E = Extroversion; A = Agreeableness; C = Conscientiousness.

4.3. *Human personality*

The analysis performed on descriptions of human personality revealed the classic five-factor structure hypothesized by the Big Five Model. Table 4 presents the main results of this rotation. Correlations among the five factors ranged from 0.17 to 0.28. Unlike descriptions of brand personality, in this case the five factors suggested by the scree test were substantially similar to those hypothesized by the model. The correctness of this interpretation is confirmed when considering the correlations of factor scores derived from the pattern matrix in Table 4 and the scores in the Big Five computed by the simple summation of the eight adjectives related to each factor (this index is called “scale validity coefficient” – see Cattell & Tsujioka, 1964). The correlations among factors and theoretical a priori scores were very high for all factors, being 0.98 for Factor 1 (Emotional Stability), 0.98 for Factor 2 (Openness), 0.91 for Factor 3 (Extroversion), 0.95 for Factor 4 (Agreeableness) and 0.81 for Factor 5 (Conscientiousness).

4.4. *Replicability of factor structure (Everett's method)*

Results of replicability analysis were clear: when exactly two factors were extracted very high replicability coefficients were found, all above 0.96. Values of replicability coefficients over 0.90 are usually considered as indicating a good level of correspondence among factors or components (Ten Berge, 1986). When more than two factors were extracted, unreplicable factors appeared in all analyses. By Everett's criterion, no more than two factors are replicable across relatively large and comparable samples.

5. Conclusions

Analyses made on descriptors of the principal dimensions of human personality revealed that the “penta-factorial” structure hypothesized by the Big Five Model was fully replicated only when subjects described their own personality. When considering brand personalities the structures that emerged were substantially divergent from the Big Five Model, being made up of two broad factors that are blends of the Big Five. In the case of brands, while exploratory factor analysis and replicability analysis clearly support a two-factor solution for brand personality descriptions, the five-factor

solution is both untenable according to the scree test and unreplicable according to the Everett method.

All analyses produced a factor defined by markers of Agreeableness and Emotional Stability. This factor (defined by terms such as “patient” and “affectionate”) reflects those aspects of brands linked to stability, predictability, and pleasantness. The other factor that occurs most frequently is defined by markers of Extroversion and Openness (with terms like “active” and “modern” as markers); this factor reflects those aspects of brands linked to dynamism, activity, and innovation. Such results are in line with findings of Digman (1997), as they replicate the same meta-factors (α and β) which derive from the blends of the five dimensions at a higher level of abstraction in the hierarchical organization of personality characteristics. Other traits specific to brands and extraneous to human personality should probably be taken into account to achieve a comprehensive picture of brand personality. This argument is sustained by the findings of other studies in which adjectives related to brand notoriety, availability, convenience, functionality, and benefits (such as “economical”, “convenient”, “available”, “useful”, “easy”, “well-known”, “famous”, “great”, and “celebrated”) have been added to Big Five markers in describing brands personality (Caprara & Barbaranelli, 1996; Guido, 1997, 2001).

As noted by Aaker (1997), the best way to compile adjectives for measuring brand personality has not yet been defined. According to our findings, it is also questionable whether the same markers can be applied to all brands. In fact, the same adjectives locate under different factors not only when comparing descriptions of human and brand personalities, but also when comparing descriptions of different brands. Several markers, like “energetic”, “conscientious”, “stable”, and “creative”, shifted from one factor to another depending on the brands they were describing. Such a phenomenon is reminiscent of the phenomenon of *concept–scale interaction* described in the semantic differential approach to the study of connotative meanings of concepts (see Heise, 1969; Kubiniec & Bean, 1978; Mann, Phillips, & Thompson, 1979; Osgood, Suci, & Tannenbaum, 1957). Researchers used this label to refer to the fact that the meaning of an adjective, and its relation with other adjectives, varies according to the concept to which the adjective refers. Indeed, relationships among adjectives are not determined by the sharing of a common component independent of the concept. At the core of concept–scale interaction lies the fact that adjectives may have different degrees of relevance for different concepts. If an adjective is irrelevant to a concept, ratings on it will have low correlations with ratings on other

adjectives that are relevant to the concept, thus giving the adjective a factor location different from its usual one.

Moreover, the environment evoked by concepts may instigate semantic shifts in adjectives (i.e., changes in the meaning of words), which leads to differences in the relations among the same adjectives and hence in factor structures, depending on the concepts they are applied to. In turn, different sets of factors (differing in number and/or nature) may result depending on the concepts that are rated (Pedhazur & Pedhazur, 1991). This could be described as *brand–adjective interaction*, meaning that adjectives may assume different meanings when used to describe different brands. This implies that adjectives have a “contextual” or “relational” meaning, that is, that they convey different meanings as they move from one dimension to another according to the brand they are describing. Similar results, for example, were found in the evaluation of the personality of political leaders where the number of traits was restricted, with the same adjectives locating under different factors (Caprara, Barbaranelli, & Zimbardo, 1997).

Brand–adjective interaction must be taken into account whenever traits such as those identified within the framework of the psycho-lexical approach are used to derive descriptions of brands as well as humans. To the extent covered by our findings, it is difficult to rely on the same factorial structure, markers, or adjective meanings. Nor should one automatically assume that subjects will respond in a similar way to a series of brands simply because they belong to the same market domain (cf. Kubiniec & Bean, 1978). Researchers should empirically validate the dimensionality of any brand descriptions by means of ad hoc factor analyses, which should be performed within each single brand on the scales used to assess them. In order to avoid ambiguous, invalid or unreliable results, factor structures cannot be accepted a priori without attempting to assess the presence of an interaction between the concept to be described (for example a brand or a product) and the terms to be used to describe it. This step is crucial before comparing different brands across personality dimensions.

In conclusion, while it may be possible to describe brand personalities with only a few factors, it is unlikely that the same factors used to describe human personality are suitable for the description of brands. In this regard, our findings are in line with those of Aaker (1997, p. 348), who observed that “though some dimensions (or factors) of human personality may be mirrored in brands, others might not”. The traditional repertoire of human personality may serve for construing a brand personality, but only to a certain extent. When applied to products and brands, the Big Five Model needs revision and

adaptation. Results show that only the two meta-factors, blends of the five main dimensions, are consistently replicated in brand perceptions. Most importantly, moreover, adjectives used to describe those traits may “shift” from one factor to another, according to the type of the selected stimulus brand.

Such limitations, however, do not invalidate the use of the *psycholexical approach* as a vital tool for studying brand personality and detecting the best adjectives marketers can use to shape their desirable brand personality. This approach allows us to distinguish among the main distinctive “traits” of specific brands, and to select words and messages which may most effectively convey (and reinforce) the competitive characteristics of brands. Future research is needed to clarify how the contextual meanings of words shape brands’ personalities, and to design appropriate measures for assessing personality characteristics capable of affecting purchasing decisions.

Appendix A. Adjectives used for brand/human personality assessment

Extroversion: Active, Competitive, Dominant, Energetic, Happy, Lively, Resolute, Strong

Agreeableness: Affectionate, Altruist, Authentic, Cordial, Faithful, Generous, Genuine, Loyal

Conscientiousness: Conscientious, Constant, Efficient, Precise, Productive, Regular, Reliable, Scrupulous

Emotional Stability: Calm, Level-headed, Light-hearted, Patient, Relaxed, Serene, Stable, Tranquil

Openness: Creative, Fanciful, Informed, Innovating, Modern, Original, Recent, Up-to-date

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